BOM

Component	Material / note	Mass (g/cell)	kg/kWh	Formula	Source
Positive active material	LNMO $LiNi_{0.5}Mn_{1.5}O_4$, vol frac 0.665	20.61	0.586	coating mass × active vol frac	LNMO.json active vol fraction; calc-energy layer_kg_m2
Positive conductive additive	carbon black (lit. ρ 1900 kg/m³), est. 0.067 vol frac	2.08	0.059	coating mass × est. vol frac (lit. default)	literature default (annotated: split 80/20 of residual 0.335)
Positive binder	PVDF (lit. ρ 1780 kg/m³), est. 0.268 vol frac	8.31	0.236	coating mass × est. vol frac (lit. default)	literature default (annotated)
Negative active material	graphite, vol frac 0.75	15.62	0.444	coating mass × active vol frac	Chen2020 default active vol fraction
Negative conductive additive	est. 0.05 vol frac (lit. default)	1.04	0.03	coating mass × est. vol frac	literature default (annotated)
Negative binder	CMC/SBR, est. 0.20 vol frac (lit. default)	4.17	0.118	coating mass × est. vol frac	literature default (annotated)
Separator	default sep. ρ 397 kg/m³	0.17	0.005	L×(1–ε)×ρ×area	calc-energy layer_kg_m2
Electrolyte	HiTrans-class + FEC/VC/PES/DTD; ρ 1.2 g/cm³ (lit.)	11.33	0.322	pore volume × 1.2 g/cm³	pore vol = area×Σ(L·ε); density literature (annotated)
Positive current collector (Al)	Al 8 µm	2.22	0.063	L×ρ×area	calc-energy layer_kg_m2
Negative current collector (Cu)	Cu 5 µm	4.6	0.131	L×ρ×area	calc-energy layer_kg_m2
Enclosure / tabs	Not modeled	N/A	N/A		beyond pure simulation boundary
TOTAL (without electrolyte)		58.81	1.673	Σ layer masses (contract caliber)	calc-energy mass_kg
TOTAL (with electrolyte)		70.1	1.995	contract + electrolyte estimate	pore-volume calc
Cell energy		35.16	Wh	V·I integration	cell/r9_lnmo_r6_energy_dfn.json:energy_wh